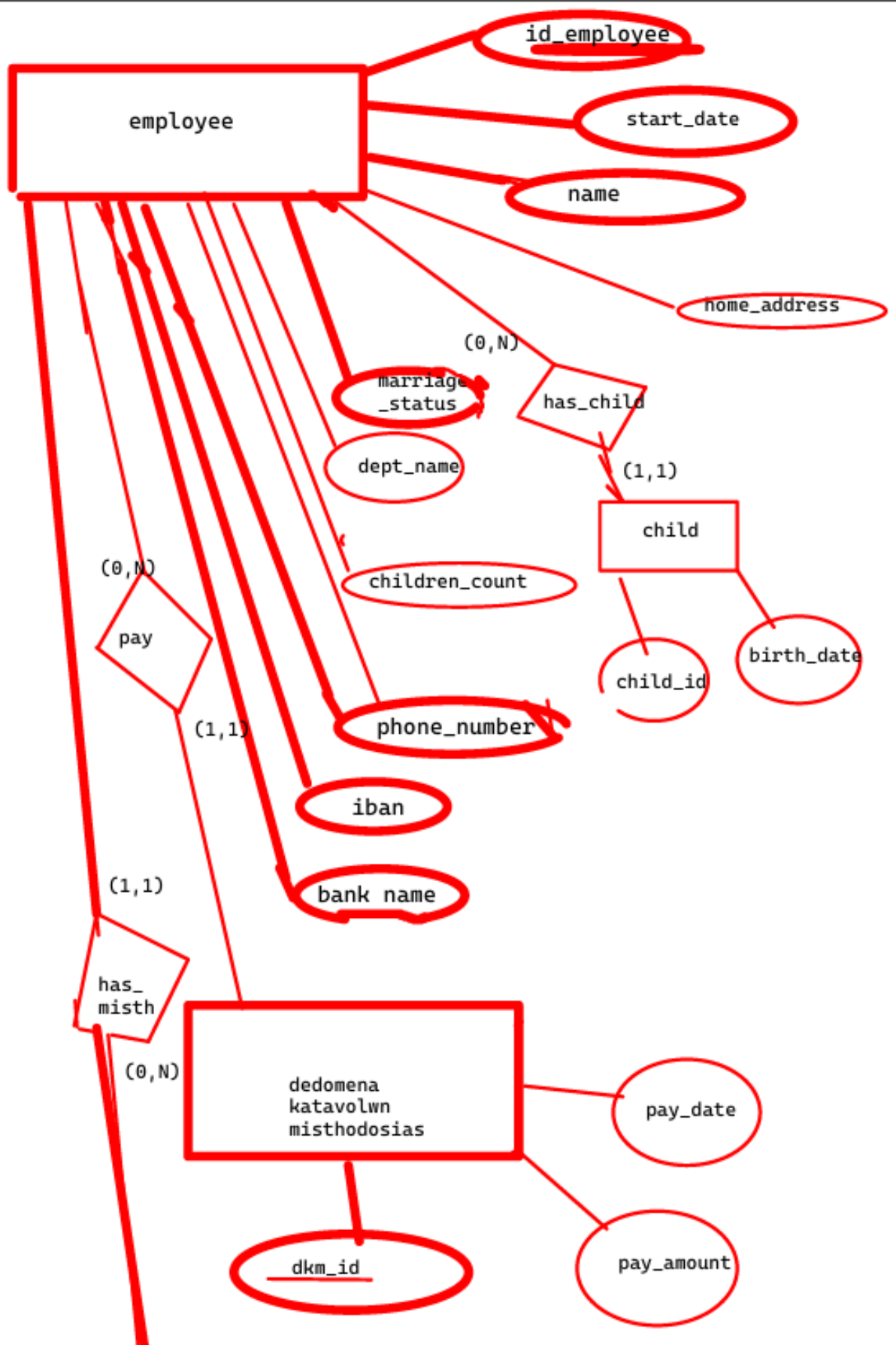
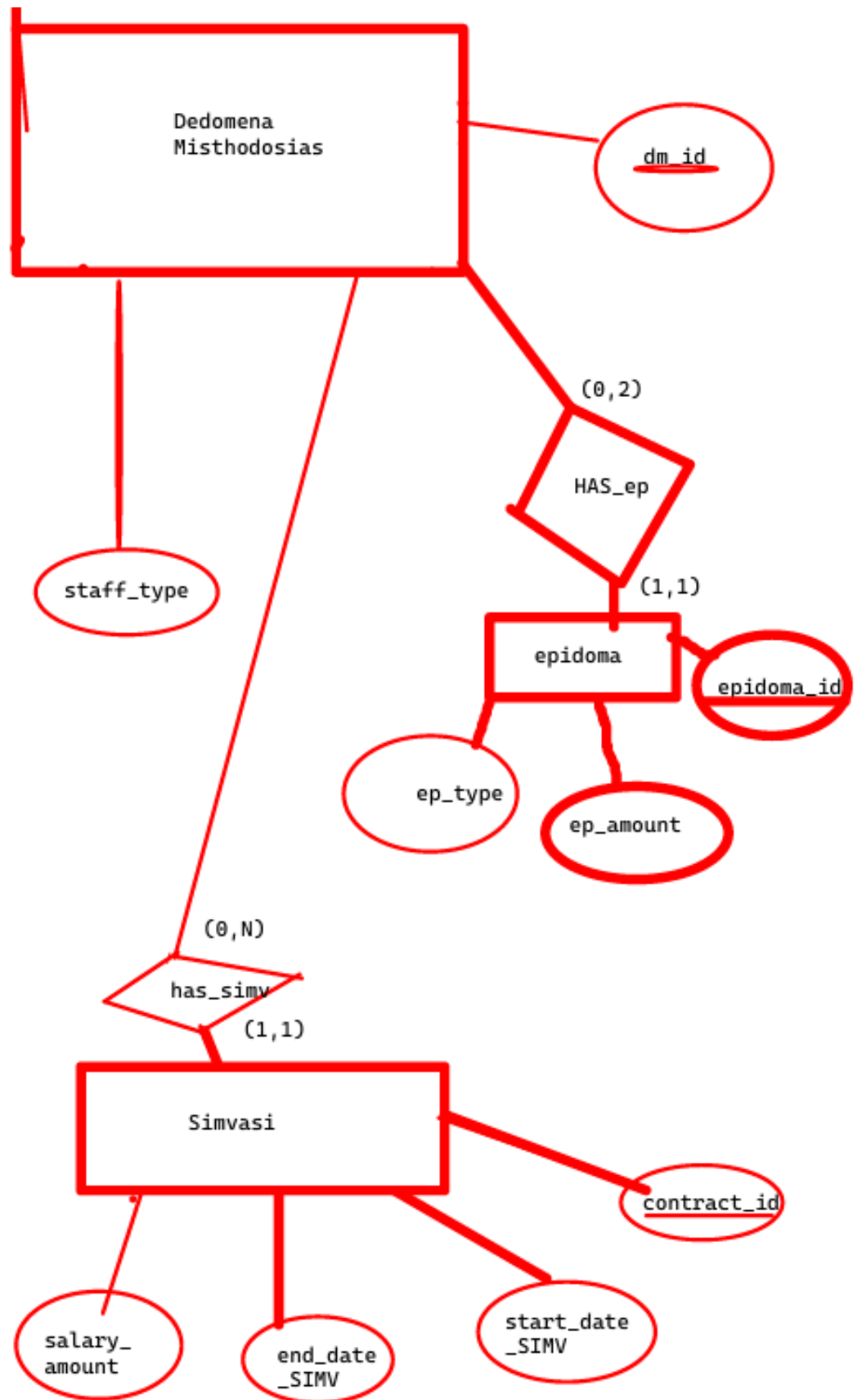






ΓΝΩΡΙΣΜΑΤΑ(ΟΝΟΜΑ ΚΑΙ ΤΥΠΟΣ)

NAME	TYPE
------	------

EMPLOYEE

id_employee	INT
name	VARCHAR(100)
phone_number	VARCHAR(100)
home_address	VARCHAR(100)
start_date	VARCHAR(100)
dept_name	VARCHAR(100)
iban	VARCHAR(100)
bank_name	VARCHAR(100)
marriage_status	VARCHAR(100)
children_count	INT

CHILD

child_id	INT
birth_date	VARCHAR(100)

HAS_CHILD

id_employee	INT
child_id	INT

DEDOMENA_KATAVOLWN_MISTHODOSIAS

dkm_id	INT
pay_date	VARCHAR(100)
pay_amount	INT

PAY

id_employee	INT
dkm_id	INT

DEDOMENA_MISTHODOSIAS

dm_id INT
staff_type VARCHAR(100)

HAS_MISTH

id_employee INT
dm_id INT

SIMVASI

contract_id INT
salary_amount INT
start_date_SIMV VARCHAR(100)
end_date_SIMV VARCHAR(100)

HAS_SIMV

dm_id INT
contract_id INT

EPIDOMA

epidoma_id INT
ep_amount INT
ep_type VARCHAR(100)

HAS_EP

dm_id INT
epidoma_id INT

ΠΡΩΤΕΥΟΝΤΑ ΚΛΕΙΔΙΑ

EMPLOYEE PRIMARY KEY

id_employee PK

CHILD

child_id PK

HAS_CHILD

id_employee PK

child_id PK

DEDOMENA_KATAVOLWN_MISTHODOSIAS

dkm_id PK

PAY

id_employee PK

dkm_id PK

DEDOMENA_MISTHODOSIAS

dm_id PK

HAS_MISTH

id_employee PK

dm_id PK

SIMVASI

contract_id PK

HAS_SIMV

dm_id PK

contract_id PK

EPIDOMA

epidoma_id PK

HAS_EP

dm_id PK

epidoma_id PK

ΠΕΡΙΟΡΙΣΜΟΙ ΠΛΗΘΙΚΟΤΗΤΑΣ

EMPLOYEE – HAS_CHILD – CHILD

Κάθε employee μπορεί να έχει 0 ή περισσότερα παιδιά (0,N)

Κάθε child ανήκει σε 1 μόνο employee (1,1)

EMPLOYEE – PAY – DEDOMENA_KATAVOLWN_MISTHODOSIAS

Κάθε employee μπορεί να έχει 0 ή περισσότερες καταβολές (0,N)

Κάθε dkm_id συνδέεται με 1 μόνο employee (1,1)

EMPLOYEE – HAS_MISTH – DEDOMENA_MISTHODOSIAS

Κάθε employee μπορεί να έχει 0 ή περισσότερους τύπους μισθοδοσίας (0,N)

Κάθε dm_id συνδέεται με 1 μόνο employee (1,1)

DEDOMENA_MISTHODOSIAS – HAS_EP – EPIDOMA

Κάθε τύπος μισθοδοσίας μπορεί να έχει 0 έως 2 επιδόματα (0,2)

Κάθε epidoma συνδέεται με 1 μόνο τύπο μισθοδοσίας (1,1)

DEDOMENA_MISTHODOSIAS – HAS_SIMV – SIMVASI

Κάθε τύπος μισθοδοσίας μπορεί να συνδέεται με 0 ή περισσότερες συμβάσεις (0,N)

Κάθε contract_id ανήκει σε 1 μόνο τύπο μισθοδοσίας (1,1)

ΠΕΡΙΟΡΙΣΜΟΙ ΑΚΕΡΑΙΟΤΗΤΑΣ

EMPLOYEE

id_employee PK

CHILD

child_id PK

HAS_CHILD

id_employee PK

child_id PK

DEDOMENA_KATAVOLWN_MISTHODOSIAS

dkm_id PK

PAY

id_employee PK

dkm_id PK

DEDOMENA_MISTHODOSIAS

dm_id PK

HAS_MISTH

id_employee PK

dm_id PK

SIMVASI

contract_id PK

HAS_SIMV

dm_id PK

contract_id PK

EPIDOMA

epidoma_id PK

HAS_EP

dm_id PK

epidoma_id PK

Ξένα Κλειδιά

HAS_CHILD.id_employee → EMPLOYEE.id_employee

HAS_CHILD.child_id → CHILD.child_id

PAY.id_employee → EMPLOYEE.id_employee
PAY.dkm_id → DEDOMENA_KATAVOLWN_MISTHODOSIAS.dkm_id
HAS_MISTH.id_employee → EMPLOYEE.id_employee
HAS_MISTH.dm_id → DEDOMENA_MISTHODOSIAS.dm_id
HAS_SIMV.dm_id → DEDOMENA_MISTHODOSIAS.dm_id
HAS_SIMV.contract_id → SIMVASI.contract_id
HAS_EP.dm_id → DEDOMENA_MISTHODOSIAS.dm_id
HAS_EP.epidoma_id → EPIDOMA.epidoma_id

Ακεραιότητα Τύπων Δεδομένων

id_employee, child_id, dkm_id, dm_id, contract_id, epidoma_id INT
children_count, pay_amount, salary_amount, ep_amount INT

ΣΥΝΑΡΤΗΣΙΑΚΕΣ ΕΞΑΡΤΗΣΕΙΣ & ΚΛΕΙΔΙΑ

EMPLOYEE(id_employee, name, phone_number, home_address, start_date,
dept_name, iban, bank_name, marriage_status, children_count)

CHILD(child_id, birth_date)

HAS_CHILD(id_employee, child_id)

DEDOMENA_KATAVOLWN_MISTHODOSIAS(dkm_id, pay_date, pay_amount)

PAY(id_employee, dkm_id)

DEDOMENA_MISTHODOSIAS(dm_id, staff_type)

HAS_MISTH(id_employee, dm_id)

HAS_SIMV(dm_id, contract_id)

SIMVASI(contract_id, salary_amount, start_date_SIMV, end_date_SIMV)

EPIDOMA(epidoma_id, ep_amount, ep_type)

HAS_EP(dm_id, epidoma_id)

ΟΙ ΣΥΝΑΡΤΗΣΙΑΚΕΣ ΕΞΑΡΤΗΣΕΙΣ ΠΟΥ ΙΣΧΥΟΥΝ:

id_employee-> name, phone_number, home_address, start_date, dept_name, iban,
bank_name, marriage_status, children_count

child_id-> birth_date

dkm_id-> pay_date, pay_amount

dm_id->staff_type

contract_id -> salary_amount, start_date_SIMV, end_date_SIMV

epidoma_id-> ep_amount, ep_type

ΚΛΕΙΔΙΑ:

EMPLOYEE :id_employee

CHILD: child_id

HAS_CHILD: (id_employee, child_id)

DEDOMENA_KATAVOLWN_MISTHODOSIAS: dkm_id

PAY: (id_employee, dkm_id)

DEDOMENA_MISTHODOSIAS: dm_id

HAS_MISTH : (id_employee, dm_id)

SIMVASI: contract_id

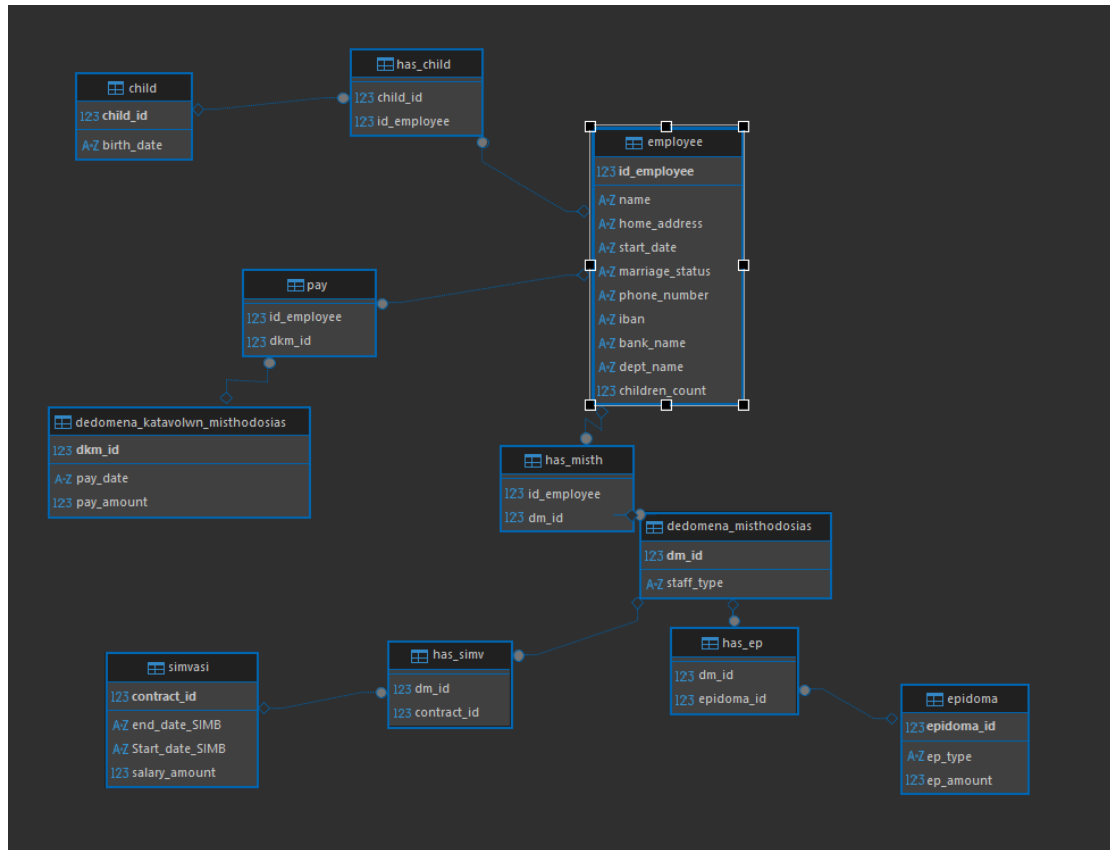
HAS_SIMV: (dm_id, contract_id)

EPIDOMA: epidoma_id

HAS_EP: (dm_id, epidoma_id)

Το δεδομένο μοντέλο είναι ήδη σε 3NF με βάση τις δοθείσες συναρτησιακές εξαρτήσεις. Δεν απαιτήθηκαν περαιτέρω διασπάσεις. Κάθε σχέση έχει πρωτεύον κλειδί καθορισμένο, όλα τα μη-κλειδιά εξαρτώνται πλήρως από το πρωτεύον κλειδί και δεν υπάρχουν μεταβατικές εξαρτήσεις.

ΣΧΕΣΙΑΚΟ ΜΟΝΤΕΛΟ



SQL:

/*Συνολικό ύψος μισθοδοσίας ανά κατηγορία προσωπικού*/

-- create view sum_Misthodosia_Dioikitikos as MD

```
select sum (dkm.pay_amount) as total_pay_per_Category
from dedomena_katavolwn_misthodosias dkm
join pay p on p.dkm_id = dkm.dkm_id
join employee e on e.id_employee = p.id_employee
join has_misth hm on e.id_employee = hm.id_employee
join dedomena_misthodosias dm on hm.dm_id = dm.dm_id
where dm.staff_type = 'Dioikitikos'
```

```
select sum (dkm.pay_amount) as total_pay_per_Category
from dedomena_katavolwn_misthodosias dkm
join pay p on p.dkm_id = dkm.dkm_id
join employee e on e.id_employee = p.id_employee
join has_misth hm on e.id_employee = hm.id_employee
join dedomena_misthodosias dm on hm.dm_id = dm.dm_id
where dm.staff_type = 'Didaktikos'
```

/* Στοιχεία και μισθοδοσία συγκεκριμένου υπαλλήλου*/

-- create view hy360_payroll_database.Employee_found as

select e.*,dkm.*

```
from hy360_payroll_database.employee e,
hy360_payroll_database.dedomena_katavolwn_misthodosias dkm ,
hy360_payroll_database.pay p
```

where p.id_employee =e.id_employee and p.dkm_id = dkm.dkm_id

and e.id_employee = find_id /* will be set by user through JAVA*/

/*Μεγιστος, ελάχιστος και μέσος μισθός ανά κατηγορία προσωπικού*/

```
SELECT dm.staff_type,
MAX(dkm.pay_amount) AS MaxPayAmount,
MIN(dkm.pay_amount) AS MinPayAmount,
AVG(dkm.pay_amount) AS AvgPayAmount
FROM hy360_payroll_database.EMPLOYEE e,
hy360_payroll_database.DEDOMENA_KATAVOLWN_MISTHODOSIAS dkm,
hy360_payroll_database.DEDOMENA_MISTHODOSIAS dm,
hy360_payroll_database.PAY p, hy360_payroll_database.HAS_MISTH m
WHERE e.id_employee = p.id_employee
AND p.dkm_id = dkm.dkm_id
AND m.id_employee = p.id_employee
AND dm.dm_id = m.dm_id
AND dm.staff_type = 'Dioikitikos'
GROUP BY dm.staff_type
```

```
SELECT dm.staff_type,
MAX(dkm.pay_amount) AS MaxPayAmount,
MIN(dkm.pay_amount) AS MinPayAmount,
AVG(dkm.pay_amount) AS AvgPayAmount
FROM hy360_payroll_database.EMPLOYEE e,
hy360_payroll_database.DEDOMENA_KATAVOLWN_MISTHODOSIAS dkm,
hy360_payroll_database.DEDOMENA_MISTHODOSIAS dm,
hy360_payroll_database.PAY p, hy360_payroll_database.HAS_MISTH m
WHERE e.id_employee = p.id_employee
AND p.dkm_id = dkm.dkm_id
AND m.id_employee = p.id_employee
AND dm.dm_id = m.dm_id
AND dm.staff_type = 'Didaktikos'
GROUP BY dm.staff_type
```

```

-- /*Κατάσταση μισθοδοσίας ανά κατηγορία προσωπικού:*/

-- create view hy360_payroll_database.SalarystatusforDioikitikos as

SELECT

e.id_employee,

e.name,

dkm.pay_amount

FROM hy360_payroll_database.EMPLOYEE e,
hy360_payroll_database.DEDOMENA_KATAVOLWN_MISTHODOSIAS dkm,
hy360_payroll_database.DEDOMENA_MISTHODOSIAS dm,
hy360_payroll_database.PAY p, hy360_payroll_database.HAS_MISTH m

WHERE e.id_employee = p.id_employee

AND dkm.dkm_id = p.dkm_id

AND m.id_employee = e.id_employee

AND dm.dm_id = m.dm_id

AND dm.staff_type = 'Dioikitikos'


-- create view hy360_payroll_database.Salary_status_for_Didaktikos as

SELECT

e.id_employee,

e.name,

dkm.pay_amount

FROM hy360_payroll_database.EMPLOYEE
e,hy360_payroll_database.DEDOMENA_KATAVOLWN_MISTHODOSIAS dkm,
hy360_payroll_database.DEDOMENA_MISTHODOSIAS dm,
hy360_payroll_database.PAY p, hy360_payroll_database.HAS_MISTH m

WHERE e.id_employee = p.id_employee

AND dkm.dkm_id = p.dkm_id

AND m.id_employee = e.id_employee

AND dm.dm_id = m.dm_id

AN D dm.staff_type = 'Didaktikos'

```

ΕΝΤΟΛΕΣ ΤΗΣ ΓΛΩΣΣΑΣ ΟΡΙΣΜΟΥ ΔΕΔΟΜΕΝΩΝ ΓΙΑ ΤΙΣ ΣΧΕΣΕΙΣ ΠΟΥ ΠΡΟΚΥΠΤΟΥΝ

```
DROP TABLE IF EXISTS `child` ;

/*!40101 SET @saved_cs_client  = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;

CREATE TABLE `child` (
  `child_id` int(11) NOT NULL AUTO_INCREMENT,
  `birth_date` varchar(100) DEFAULT NULL,
  PRIMARY KEY (`child_id`)
) ENGINE=InnoDB AUTO_INCREMENT=107 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;

/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Table structure for table `dedomena_katavolwn_misthodosias`
--

DROP TABLE IF EXISTS `dedomena_katavolwn_misthodosias` ;

/*!40101 SET @saved_cs_client  = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;

CREATE TABLE `dedomena_katavolwn_misthodosias` (
  `pay_date` varchar(100) DEFAULT NULL,
  `pay_amount` int(11) DEFAULT NULL,
  `dkm_id` int(11) NOT NULL AUTO_INCREMENT,
  PRIMARY KEY (`dkm_id`)
) ENGINE=InnoDB AUTO_INCREMENT=3932 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;

/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Table structure for table `dedomena_misthodosias`
--
```

```

DROP TABLE IF EXISTS `dedomena_misthodosias`;

/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `dedomena_misthodosias` (
  `dm_id` int(11) NOT NULL AUTO_INCREMENT,
  `staff_type` varchar(100) DEFAULT NULL,
  PRIMARY KEY (`dm_id`)
) ENGINE=InnoDB AUTO_INCREMENT=161 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--

-- Table structure for table `employee`
--

DROP TABLE IF EXISTS `employee`;

/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `employee` (
  `name` varchar(100) DEFAULT NULL,
  `home_address` varchar(100) DEFAULT NULL,
  `start_date` varchar(100) DEFAULT NULL,
  `marriage_status` varchar(100) DEFAULT NULL,
  `phone_number` varchar(100) DEFAULT NULL,
  `iban` varchar(100) DEFAULT NULL,
  `bank_name` varchar(100) DEFAULT NULL,
  `id_employee` int(11) NOT NULL AUTO_INCREMENT,
  `dept_name` varchar(100) DEFAULT NULL,
  `children_count` int(11) DEFAULT NULL,
  PRIMARY KEY (`id_employee`)

```

```
) ENGINE=InnoDB AUTO_INCREMENT=191 DEFAULT CHARSET=utf8mb4  
COLLATE=utf8mb4_general_ci;
```

```
/*!40101 SET character_set_client = @saved_cs_client */;
```

```
--
```

```
-- Temporary view structure for view `employee_found`
```

```
--
```

```
DROP TABLE IF EXISTS `employee_found`;
```

```
/*!50001 DROP VIEW IF EXISTS `employee_found` */;
```

```
SET @saved_cs_client = @@character_set_client;
```

```
/*!50503 SET character_set_client = utf8mb4 */;
```

```
/*!50001 CREATE VIEW `employee_found` AS SELECT
```

```
1 AS `name`,
```

```
1 AS `home_address`,
```

```
1 AS `start_date`,
```

```
1 AS `marriage_status`,
```

```
1 AS `phone_number`,
```

```
1 AS `iban`,
```

```
1 AS `bank_name`,
```

```
1 AS `id_employee`,
```

```
1 AS `dept_name`,
```

```
1 AS `pay_date`,
```

```
1 AS `pay_amount`,
```

```
1 AS `dkm_id` */;
```

```
SET character_set_client = @saved_cs_client;
```

```
--
```

```
-- Table structure for table `epidoma`
```

```
--
```

```

DROP TABLE IF EXISTS `epidoma`;

/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `epidoma` (
  `epidoma_id` int(11) NOT NULL AUTO_INCREMENT,
  `ep_type` varchar(100) DEFAULT NULL,
  `ep_amount` double DEFAULT NULL,
  PRIMARY KEY (`epidoma_id`)
) ENGINE=InnoDB AUTO_INCREMENT=321 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--

-- Table structure for table `has_child`
--

DROP TABLE IF EXISTS `has_child`;

/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `has_child` (
  `child_id` int(11) DEFAULT NULL,
  `id_employee` int(11) DEFAULT NULL,
  KEY `has_child_child_FK` (`child_id`),
  KEY `has_child_employee_FK_1` (`id_employee`),
  CONSTRAINT `has_child_child_FK` FOREIGN KEY (`child_id`) REFERENCES `child`
  (`child_id`),
  CONSTRAINT `has_child_employee_FK` FOREIGN KEY (`id_employee`) REFERENCES
  `employee` (`id_employee`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--

```

-- Table structure for table `has\_ep`

--

DROP TABLE IF EXISTS `has\_ep`;

/*!40101 SET @saved_cs_client = @@character_set_client */;

/*!50503 SET character_set_client = utf8mb4 */;

CREATE TABLE `has\_ep` (

 `dm\_id` int(11) DEFAULT NULL,

 `epidoma\_id` int(11) DEFAULT NULL,

 KEY `has\_ep\_dedomena\_misthodosias\_FK` (`dm\_id`),

 KEY `has\_ep\_epidoma\_FK` (`epidoma\_id`),

 CONSTRAINT `has\_ep\_dedomena\_misthodosias\_FK` FOREIGN KEY (`dm\_id`) REFERENCES `dedomena\_misthodosias` (`dm\_id`),

 CONSTRAINT `has\_ep\_epidoma\_FK` FOREIGN KEY (`epidoma\_id`) REFERENCES `epidoma` (`epidoma\_id`)

) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

/*!40101 SET character_set_client = @saved_cs_client */;

--

-- Table structure for table `has\_misth`

--

DROP TABLE IF EXISTS `has\_misth`;

/*!40101 SET @saved_cs_client = @@character_set_client */;

/*!50503 SET character_set_client = utf8mb4 */;

CREATE TABLE `has\_misth` (

 `id\_employee` int(11) DEFAULT NULL,

 `dm\_id` int(11) DEFAULT NULL,

 KEY `has\_misth\_employee\_FK` (`id\_employee`),

 KEY `has\_misth\_dedomena\_misthodosias\_FK` (`dm\_id`),

 CONSTRAINT `has\_misth\_dedomena\_misthodosias\_FK` FOREIGN KEY (`dm\_id`) REFERENCES `dedomena\_misthodosias` (`dm\_id`),

```

    CONSTRAINT `has_misth_employee_FK` FOREIGN KEY (`id_employee`)
REFERENCES `employee` (`id_employee`)

) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

/*!40101 SET character_set_client = @saved_cs_client */;

--

-- Table structure for table `has_simv`

--

DROP TABLE IF EXISTS `has_simv`;

/*!40101 SET @saved_cs_client = @@character_set_client */;

/*!50503 SET character_set_client = utf8mb4 */;

CREATE TABLE `has_simv` (
  `dm_id` int(11) DEFAULT NULL,
  `contract_id` int(11) DEFAULT NULL,
  KEY `has_simv_dedomena_misthodosias_FK` (`dm_id`),
  KEY `has_simv_simvasi_FK` (`contract_id`),
  CONSTRAINT `has_simv_dedomena_misthodosias_FK` FOREIGN KEY (`dm_id`)
REFERENCES `dedomena_misthodosias` (`dm_id`),
  CONSTRAINT `has_simv_simvasi_FK` FOREIGN KEY (`contract_id`) REFERENCES
`simvasi` (`contract_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

/*!40101 SET character_set_client = @saved_cs_client */;

--

-- Table structure for table `pay`

--

DROP TABLE IF EXISTS `pay`;

/*!40101 SET @saved_cs_client = @@character_set_client */;

/*!50503 SET character_set_client = utf8mb4 */;

CREATE TABLE `pay` (

```

```

`id_employee` int(11) DEFAULT NULL,
`dkm_id` int(11) DEFAULT NULL,
KEY `pay_employee_FK` (`id_employee`),
KEY `pay_dedomena_katavolwn_misthodosias_FK` (`dkm_id`),
CONSTRAINT `pay_dedomena_katavolwn_misthodosias_FK` FOREIGN KEY
(`dkm_id`) REFERENCES `dedomena_katavolwn_misthodosias` (`dkm_id`),
CONSTRAINT `pay_employee_FK` FOREIGN KEY (`id_employee`) REFERENCES
`employee` (`id_employee`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Temporary view structure for view `salary_status_for_didaktikos`
--

DROP TABLE IF EXISTS `salary_status_for_didaktikos`;
/*!50001 DROP VIEW IF EXISTS `salary_status_for_didaktikos` */;
SET @saved_cs_client = @@character_set_client;
/*!50503 SET character_set_client = utf8mb4 */;
/*!50001 CREATE VIEW `salary_status_for_didaktikos` AS SELECT
1 AS `id_employee`,
1 AS `name`,
1 AS `pay_amount` */;
SET character_set_client = @saved_cs_client;

--
-- Table structure for table `simvasi`
--

DROP TABLE IF EXISTS `simvasi`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;

```

```

CREATE TABLE `simvasi` (
  `contract_id` int(11) NOT NULL AUTO_INCREMENT,
  `end_date_SIMB` varchar(100) DEFAULT NULL,
  `Start_date_SIMB` varchar(100) DEFAULT NULL,
  `salary_amount` int(11) DEFAULT NULL,
  PRIMARY KEY (`contract_id`)
) ENGINE=InnoDB AUTO_INCREMENT=72 DEFAULT CHARSET=utf8mb4
COLLATE=utf8mb4_general_ci;

/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Temporary view structure for view `sum_misthodosia_didaktikos`
--

DROP TABLE IF EXISTS `sum_misthodosia_didaktikos`;

/*!50001 DROP VIEW IF EXISTS `sum_misthodosia_didaktikos` */;

SET @saved_cs_client = @@character_set_client;

/*!50503 SET character_set_client = utf8mb4 */;

/*!50001 CREATE VIEW `sum_misthodosia_didaktikos` AS SELECT
  1 AS `sum(dkm.pay_amount)` */;

SET character_set_client = @saved_cs_client;

```

